# Social Control and Walkability: Evaluating Jaipur's Urban Design and Pedestrian Mobility

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## 1. Introduction

Walking is the human civilisation's oldest mode of mobility and today this silent act shapes how cities are lived, perceived, and shared. Walkability as a concept signifies that urban spaces are designed to integrate walking as viable and desirable mode of transportation. Walkability can be viewed as the nexus for three interconnected goals of health, sustainability, and liveability (Baobeid et. al., 2021). Walking, as a leisure activity or as a form of transportation, improves physical and mental health; aligns with global climate goals of reducing carbon emissions and; makes neighbourhoods socially cohesive. Thus, the concept of walkability transcends mere infrastructure as it is deeply entrenched in the broader vision of achieving equitable and resilient urban environment. So, what makes for a "good" walk? Jeff Speck's "General Theory of Walkability"[1] aims that the following four pillars should coexist – proximity to destinations (reason to walk / useful); safety (actual and perceived); comfort and; interest (TEDx Talks, 2014). The first pillar highlights that a walk should serve a purpose – places to walk to- and thus requires connectivity, mixed-use development of land and "15-minute cities"[2]. Next, safety for pedestrians should be from both traffic and crime. Further, for comfort, streetscapes should be designed at human scale with appropriate shade, seating, and elevated pathways. The last pillar is that the walk should be interesting and could be achieved with a rich and active environment that fosters a stimulating experience.

Despite the recognised benefits improving walkability holds. It remains a severely overlooked aspect in urban planning in India. Cities fail to cohesively integrate pedestrian movement while prioritising vehicular movement. Usage of non-motorised transport in urban cities becomes a rarer phenomenon. Poor connectivity, neglect of safety and overly utilitarian designs create a disconnect between policy in theory and realities on ground. The paper aims to analyse and address these gaps by identifying some of the systemic barriers to walkability in policy, design, and community level. With the use of GIS (Geographic Information System) a comparative case study is done in various areas of the city of Jaipur to evaluate walkability success metrics and obtain localised spatial analysis. The disjuncture between transport infrastructure and

[1] Jeff Speck is an American writer, city planner and a walkability advocate. His theory highlights the four elements for walkability (mentioned above) together make walking a desirable mode of transportation, leading to multi-fold benefits to walkable cities.

[2] "15-minute city" is an urban planning concept that advocates that essential services (like work, commerce, education, healthcare, school, park, entertainment) should be accessible to residents of a city within a 15-minute walkable or bikeable distance.

pedestrian access is highlighted via a mixed-method approach that combines spatial mapping, on-ground surveys, and observations and, scholarly literature. Walkability is situated as a core element of just mobility planning by analysing power dynamics on streets. It is intended to contribute to urban policy discourse by identifying spatial gaps and governance failures limiting walkability in Jaipur. The paper aims to underscore the need for enhanced walkability both for policymakers and citizens for building healthier, sustainable, and inclusive cities.

## 2. Jaipur's Mobility Mosaic

Jaipur, the capital city of the country's largest state, Rajasthan is one of the earliest planned cities in the country and now a national and international tourist destination. The city enjoys robust connectivity through road, rail and air, with India's vital transportation corridors running through it. There are two public transportation systems in the city – the bus network of Jaipur Bus administered by JCTSL[3] and; a metro rail system operated by JMRC[4]. Apart from the formal public transport system, Jaipur's informal modes of transport are supported with mini buses, *Tata Magics*, auto-rickshaws, *Vikrams*, jeeps and cycle rickshaws (Kumar et. al., 2016). These modes have become vital in peri-urban areas of the city where the BRTS and the metro falls short. Multiple studies (Kumar et. al., 2016; Pareek and Mathur, 2022) argue that though these transport options lack short on comfort, these are the only options for many commuters. Thus, Indian cities cannot simply dismiss them due to their negative reputation of safety as the research showed that these perceptions may not be entirely true. Further, they have rated Jaipur's urban transportation system as unsustainable while highlighting no specific targets for pedestrian movement and walking networks are defined.

While there exists no Greenhouse Gases (GHG) emission monitoring system for continuous measurement and the transport sector is the second highest GHG emitter in the city, around 29% of the total trips in the city are on foot (UN Habitat, 2021). The average home-based trip length by walking is 1.8 kms (Comprehensive Mobility Plan, Jaipur city, Vol 1, 2018). Considering these findings, prioritising walking becomes even more imperative. Jaipur Development Authority (JDA)[5] in its study highlights the following critical walkability challenges that acts as hindrances in JDA's 20-year investment roadmap for the city's transportation strategy (HT Correspondent, 2019). There is a severe dearth of pedestrian footpaths with 80% of roads in the city without them and 62% roads are encroached with on-street parking. Unsafe crossings, lack of last-mile connectivity, missing road markings and signboards and inadequate water drainage especially during monsoons are among other key

[3] The Jaipur City Transport Services Limited (JCTSL) "operates, controls and regulates" the public bus system in the city. it offers 200 standard and low-floor buses which are AC and non-AC, on 27 urban/sub-urban bus routes. It is operated under the Rajasthan State Road Transport Corporation (RSRTC). https://transport.rajasthan.gov.in/content/transportportal/en/jaipur-bus/The-Company/about-us.html

[4] The Jaipur Metro Rail Corporation Ltd. (JMRC) is the public metro rail project authority under the state government to provide "mass rapid transit system" in the city. https://transport.rajasthan.gov.in/content/transportportal/en/metro/Aboutus/Mission.html

[5] JDA is the statutory organisation under the Department of Urban Development and Housing, Government of Rajasthan. It is responsible for the city's urban development, while addressing rapidly growing population's demands and managing the spatial expansion. https://jda.rajasthan.gov.in/content/raj/udh/jda---jaipur/en/home.html#

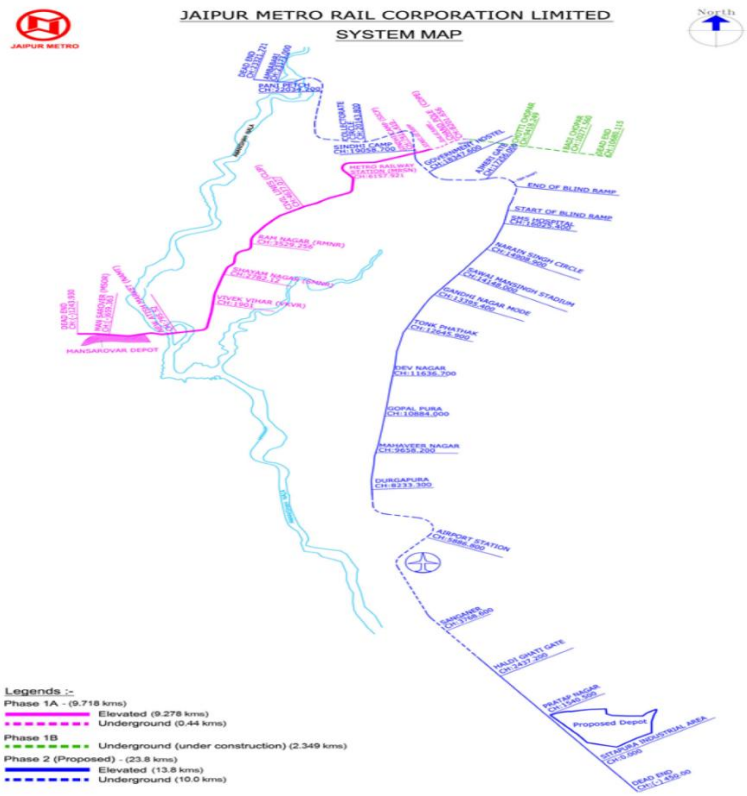
issues in the city. OMI Foundation's Ease of Moving Index reports that people's perceptions of adequacy of footpath conditions and cycle tracks have reduced from 2018 to 2022. JMRC's Comprehensive Mobility Plan of 2010 had emphasised on non-motorized transport with a target of a 100% walkability index, along with key focus on footpaths, intersections, bicycle plans, and mobility corridors. After a decade and half, we are yet to achieve these goals.

## 3. Infrastructure Failures: Metro without Momentum

Commercial operations of Phase I-A (Mansarovar to Chandpole) of the metro project commenced in 2015 and for Phase 1-B (Chandpole to Badi Chaupar) in 2020. With a total cost of Rs 3149 crores, this East-West corridor of the metro project covers 12 kms with 11 stations. The Phase II (Ambabari to Sitapura) of 23 kms, as the North-South corridor, is yet to even begin. The need for a metro rail system in the city was based on the growing population of the city. JMRC's Comprehensive Mobility Plan talks of relocation of the city's employment and commercial hubs towards the periphery areas by developing mass transit corridors, especially in North-South and East-West corridors. Mehta and Kumar (2021) highlight that though it was believed with the mass-transit system in place, it would lead to a reduction in road accidents[6], in number of vehicles owned in the households living near the station area, in pollution and, in travel time. However, none of these have materialised as the ridership is extremely low and has been on the decline after its initial year. Moreover, the Phase I offers a 25-minute travel time, without a feeder system[7], which is similar to other modes of transportation, making commuters to opt for alternatives.

[6] Mehta and Kumar (2021) reveal that though there has been a marginal decline in road accidents after 2015 (after commencement of the metro), this is due to road safety awareness.

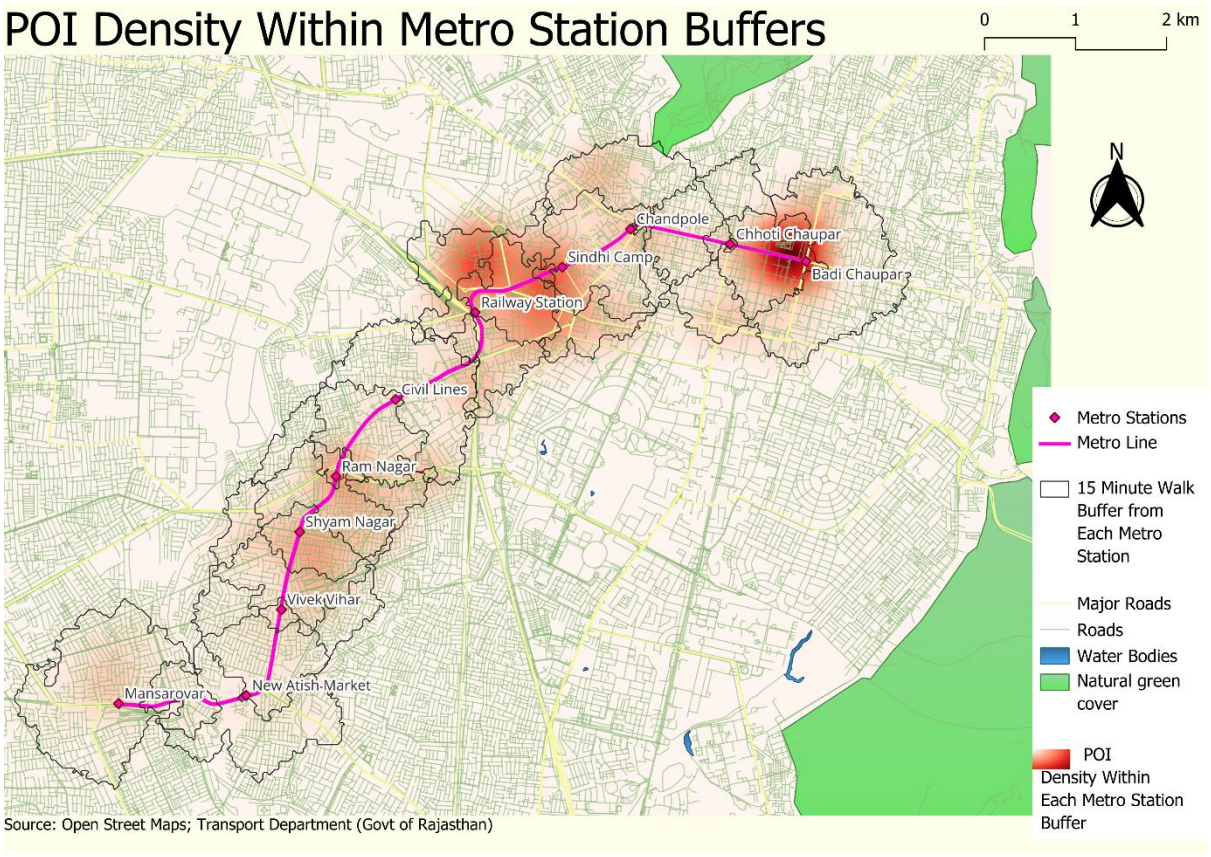
[7] Feeder services are referred to transport systems that offer short-distance travel connecting local neighbourhoods and even peripheral areas to major transport hubs like metro stations, bus terminals and railway stations. These include e-rickshaws, autos, mini-buses, vans, tempo, shared cycles, etc.




Source: Transport Department, Government of Rajasthan[8]

Map 1 marks a 15-minute walk radius from each metro station to show that it currently reaches only a fraction of the city's area and population. The current metro line leaves peripheral areas, informal settlements and middle-income zones which further reinforces mobility-based inequalities. Even if the sparse network coverage is aimed to be solved through the Phase II of the metro project, ground realities reveal that there are gaps in first and last-mile connectivity. Most metro stations lack feeder systems or presence of adequate walkways that are shaded and not obstructed.

[8] https://transport.rajasthan.gov.in/content/transportportal/en/metro/Project/Routmap.html




Map 1: POI Density Within Each Metro Station Buffers

**Key takeaways**

- Stark contrast between POI density around the metro areas
- Chandpole, Sindhi Camp stations have high POI density as they act as acitivity hubs
- Mansarovar and New Atish Market stations signal poor transit-oriented development and remain underutilised
- High POI density stations need robust feeder services on a priority basis

The metro stations also exhibit contrast in Point-of-Interest (POI) density[9] within their respective 15-minute buffer area. High POI density metro stations like Sindhi Camp (a major bus stand), Railway Station (main railway junction) and Badi Chaupar (historic walled city's commercial zone) are anchored in the city's commercial core with a transit nexus, potentially amplifying footfall. On the other hand, low POI stations like New Aatish Market (localised market) and Mansoravar (primarily a residential area) serve residential needs. However, high POI stations

[9] The POIs sourced from Open Street Maps range from hospitals, parks, shops, apartments, houses, other infrastructure, hotels, junctions etc. However, it may miss some POIs. In Map, a heatmap has been created with POIs lying in each metro buffer area to showcase variations in POI density in different metro buffer areas.

face severe congestion during peak hours, festivals, or gatherings. This coupled with narrow sidewalks, mixed traffic and vendor encroachment further intensify the discomfort. JDA's study highlights that pedestrian signals at intersections and mid-blocks does not practically exist at all at Sindhi Camp and Railway Junction (HT Correspondent, 2019). Whereas, low POI stations lack integrated pedestrian pathways. Thus to make an effective metro system focus should be towards building robust first and last-mile connectivity with shaded footways, reliable feeder services and safe pedestrian infrastructure that seamlessly integrates in mobility infrastructure.

### 3.1 Indian Metros: Missing the Last Mile

As Indian cities increasingly invest in metro infrastructure to enhance mobility, issues of actual accessibility, walkability and inclusivity still linger. The Quint's video "How Indian Cities Failed Public Transport" highlights that metro have not largely work for Indian cities because of the spatial design which is radial and thus the metro is able to offer only short trips, unless a very large and well-connected metro network is established like in Delhi. The local works well in Mumbai since the city's spatial design is lateral. Moreover, The Quint emphasises that the number of private vehicles has increased over the years and consequently congestion due to dependence on motorised means. However, the number of buses in the public transportation system have not increased accordingly. Thus, we are in the need of a very elaborate and a high-capacity public transportation system. Along with this, passenger mobility should be focussed on instead of vehicular mobility.

The goal of transit-oriented development has remained unrealized. Without well-integrated public transport system in the urban design, metro systems fail to build walkable micro-neighbourhoods. Though intended to utilise metro corridors through mixed-use areas, green spaces and public facilities, the reality remains different with a lot of the land near metro stations underutilized or commercially monopolized. Though metro systems focus on vertical mobility (connecting long distances within a city), they have been not successful in robust horizontal mobility – accessible and walkable local linkages. With lack of feeder services, the metros remain isolated axes of movement as evident in Jaipur. Thus, walkability seems to form a systemic blind spot, since huge investments are made in heavy infrastructures like metro but low-cost, high-impact pedestrian movement is neglected. Walkability should not be a by-product but deliberately designed into the urban system.

While POI density around metro stations brings out disparities in pedestrian, the case of walled city's walkability is rooted in its 18$^{th}$ century design. Let us delve deeper as the walled city acts as a historic benchmark as well as a contemporary case study.

## 4. From Vision to Vulnerability: Walled City's Walkability

The walled city was conceived as a well-planned architectural marvel of urban design in 1727 by Maharaja Sawai Jai Singh II. The urban structure brings together elements of the Vastu

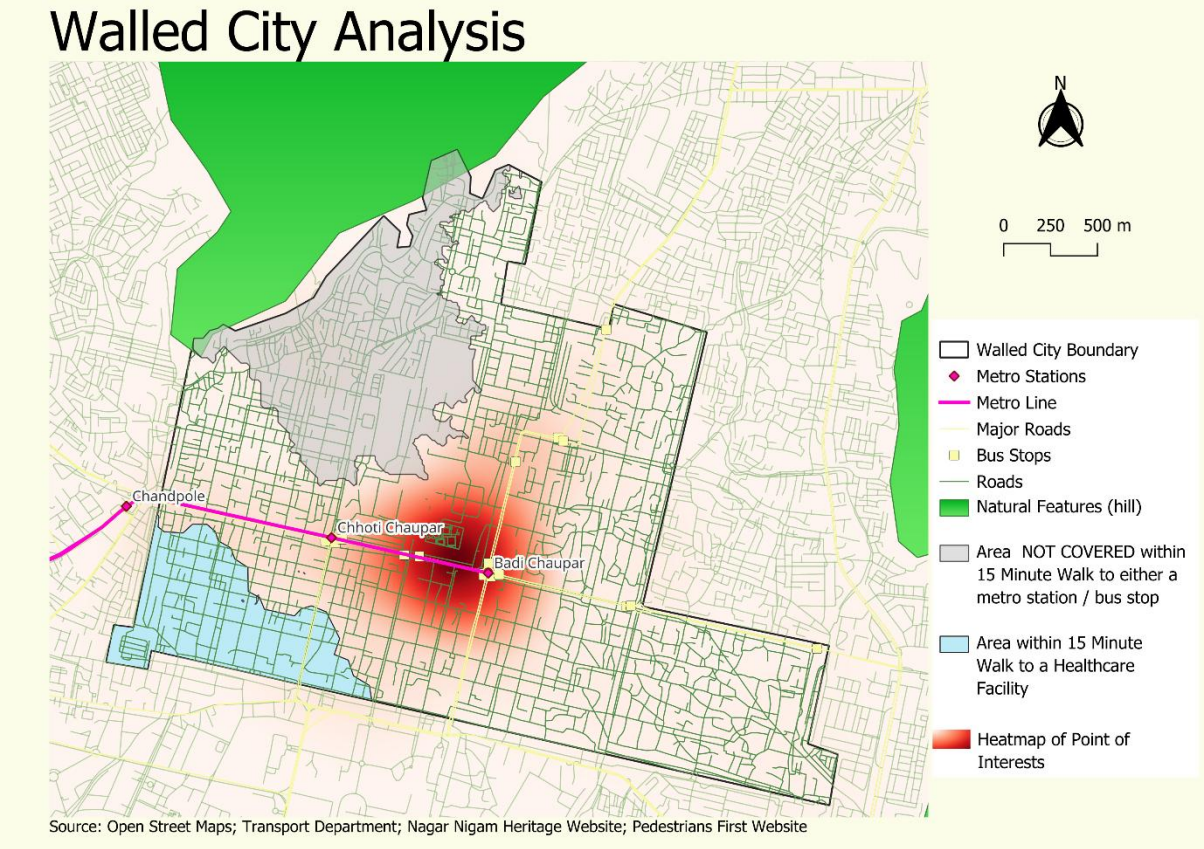
Purusha Mandala[10], using the 9 square grid (*chowkris*) as the basis of planning (Sharma, 2018). The planning of the walled city has used several principles of design such as organization of built environments to host together multiple features; balance of the streets; symmetry of the structures; order; imagery of the structure pointing towards function of the space; rhythm in patterns; spatial sequence and; circulation patterns that encourage diverse interactions, which can be different during various times of the day (Sharma et al., 2015). Closed with a boundary wall with 9 gates, the city was planned with a hierarchy of street system (main roads, secondary streets, internal streets, and local streets), facades, bazaars, mohallas, havelis, chaupars (public squares) and chowks (Pipralia et. al., 2016).

However, the contemporary development of the urban paradigm has induced strain on the visionary blueprint of the city where the walkable harmony is now under motorized chaos. Front spaces of the commercial shops on the ground floor formed a walkway for pedestrians, with roof of the first floors offering shade, easing pedestrian flow. Originally designed for non-motorised transport and pedestrian movement with open spaces (such as, Jaleb Chowk) and water bodies (such as, Talkatora), the city enhanced walkability through functional cohesion. Sharma (2018) highlights that the chaupars and gates have transitioned from impressive pedestrian zones to traffic-clogged islands. Despite wide streets (around 30 metres wide), today chaotic traffic and unorganised parking squeeze out pedestrians. Sidewalks are often obstructed or non-existent, impacting safety. The situation is aggravated with weaknesses in public transportation system, with increase in private vehicles thus worsening contestation. Limited transit options discourage walking as a viable last-mile solution.

Gadru (2017) investigates how pedestrian movement in public spaces is shaped by visual accessibility, spatial design, and climatic conditions. It reveals that pedestrian flow at Badi Chaupar significantly declines at night time due to improper street lighting, which is limited to shopfronts and a central fountain. Seating also remains absent with pedestrians resorting to fences and shop footpaths. Absence of greenery (a softscape) further diminishes aesthetic appeal. This also aggravates daytime discomfort discouraging pedestrian flow despite daytime market activity. Gestalt Principles[11] imply that pedestrians navigate public spaces based on visual coherence and pedestrian movement correlates with activity nodes such as the markets, monuments, etc. and path visibility (Gadru, 2017).

[10] In traditional Indian architecture system, Vastu Shastra, it is a geometric diagram representing a cosmic energy grid that promotes positive energy flow. Layouts and designs of buildings are thus structured accordingly.

[11] Also known as, Gestalt laws of perpetual organization, these are a set of laws that explain how human mind views, perceives, and organizes visual elements. These include laws on similarity, proximity, closure, figure-ground, continuity, symmetry, common fate, and simplicity.




Map 2: Walled City Analysis

**Key takeaways**

- Northern walled city is a transit desert
- Except for a small portion, most of the walled city lacks a healthcare facility within a 15-miute walk
- Presence of metro stations and bus stops near high POI density

The blue shaded area, in Map 2, depicts the area within a 15-minute walk to a healthcare facility[12], which only covers the southwest part of the walled city, highlighting that most of the other areas are under-served in terms of accessible healthcare by foot. The grey-shaded area represents region of the walled city not covered within a 15-minute walk from either a bus stop or a metro station. This is the northwestern part of the area, making it least walkable for transit-dependent populations. As highlighted above, the peripheral north zone thus lacks 15-minute walk access to both transit and healthcare.

[12] Data and map for a 15-miute walking access to a healthcare facility sourced from the Pedestrians First website https://pedestriansfirst.itdp.org/city-tool/step-1

The map also shows a heatmap of POI density within the walled city wherein there is highest density around the two metro stations (Chhoti Chaupar and Badi Chaupar), highlighting it as a hub of activity. Coupled with dense street grid, it implies affine-grained connectivity. The footpaths at Johari Bazaar and Bapu Bazaar are ranked as the best within the city (Vashishtha, 2023). It demonstrates how traditional architecture has accommodated pedestrian comfort with integrated and shaded sidewalks. This showcases the high potential the walled city possesses for emerging as a model of walkable heritage urbanism. In fact, Kannan (2024) credits that the concept of "15-minute" cities is not new to Indian cities and the Walled City of Jaipur already boasts of a "15-minute" city.

### 4.1 Walkability as Temporal Justice

Marc Augé describes modern urban spaces as "non-places" which symbolise anonymity, transition and as a functional zone which lack an identity. These "non-places" prioritize speed over experience and thus focus on seamless flow with minimal seating, lack of thresholds, open-plan layouts, etc. Spaces like corporate lobbies or shopping malls create a suspended sense of time. Walkability is thus reduced to a mere movement rather than a social experience. On the other hand, traditional Indian architecture has integrated in-between spaces for time to "gather." Features like jalis, courtyards, verandas, steps and thresholds create pause and presence where walkability alternates between movement and rest. While modern "non-spaces" enable physical mobility, it erodes quality of mobility.

The walled city's urban form with the presence of courtyards, shaded arcades and chaupars were originally designed as pedestrian plazas with seating and the shaded footways along the shops to foster walkability as a social experience. However, in contemporary times, cities are designed with spaces where "doing nothing" is a flaw. Streets are not just mere pathways for movement in India; they serve as multi-dimensional spaces. This is especially true for the walled city where during festivals like Gangaur and Holi, the streets transform into processional routes and during Eid into open-air praying grounds. The streets also act as economic engines with mixed-use vibrancy. Places like Kishanpole Bazaar merge workshops with retail that range from textiles to gems. Thus streets, like Johari Bazaar and Tripolia Bazaar act as "third places" – concept coined by Ray Oldenburg[13] – that are neither home nor workplace, but epicentres of informal interaction. Mehta and Bosson (2010) exemplify four physical characteristics to align with third places. These very well apply to walled city's streets as: a) *personalization* – with shopfront aesthetics, cultural markers like torans, annual whitewashing with the iconic pink paint; b) *permeability* – with open-air workshops, display windows at jewellery shops, etc.; c) *seating* – with improvised seating near raised platforms and; d) *shelter* – with covered verandas and arcades and jali screens. Notably the walled city does not host any major accident hotspot despite extremely high intersection density (covered in the next section). This reflects that a careful alignment of form and function can well serve purpose instead of modern transport technology. Preservation and adaption of traditional spatial

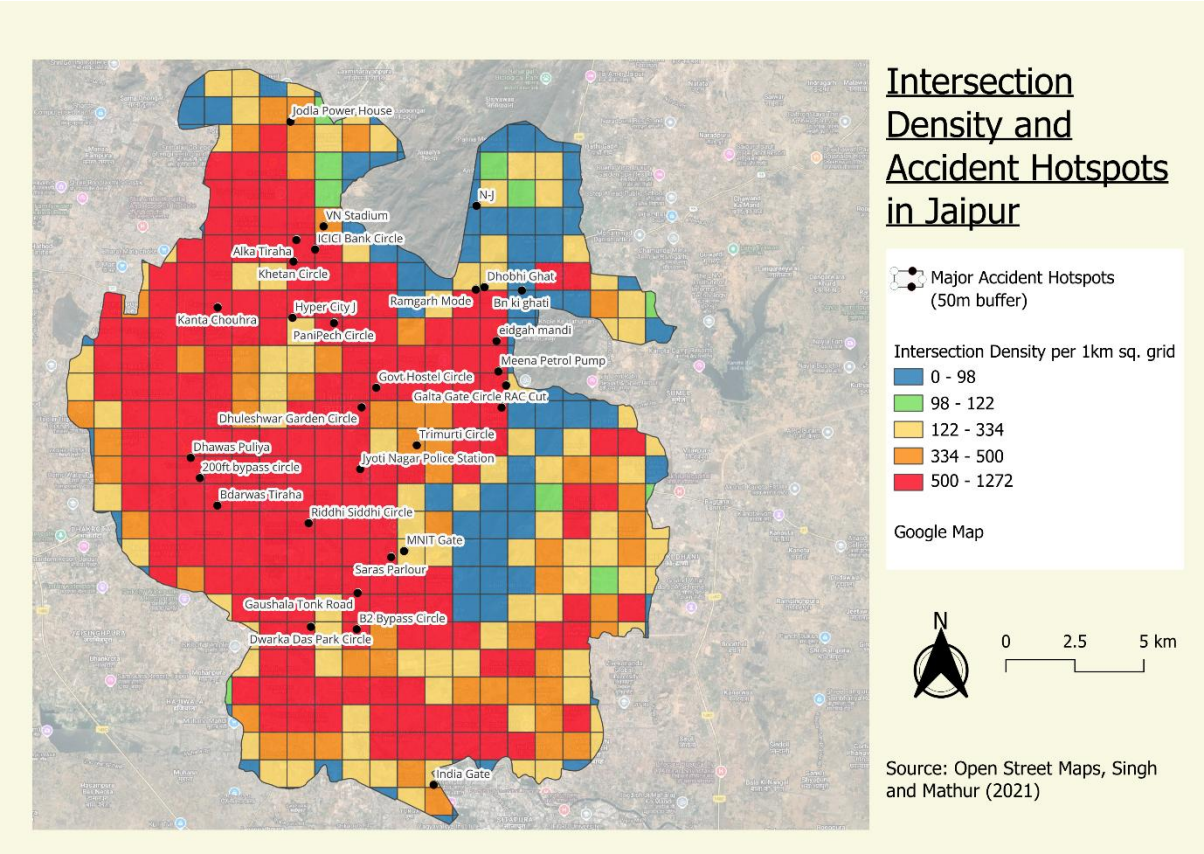
[13] An American sociologist.

elements of the city's architecture can humanise mobility fostering walkability as a rich social experience.

## 5. The Geography of Risk: Intersection Density and Accident Hotspots

Among various urban design metrics, Ewing and Cervero (2010) in their study found that intersection density holds the most important influence on pedestrian activity. It is surprising that it surpasses factors like access to public transportation, proximity to commercial locations, population density or job concentration.

Intersection density is thus considered as a good measure to assess walkability for a region. For a 1km square grid, around 100 intersections are considered ideal, with a direct relationship between the number of intersections and walkability (Practical Cities, 2021). More intersections usually translate to more connectedness which allows permeable street networks.




Map 3: Intersection Density and Accident Hotspots in Jaipur

When intersection density (per 1km square grid) and major accident hotspots[14] were chalked out for Jaipur, we see a major chunk of accident hotspots covered in very high intersection

[14] Data for major accident hotspots in the city is taken from Singh and Mathur (2021).

Key takeaways:

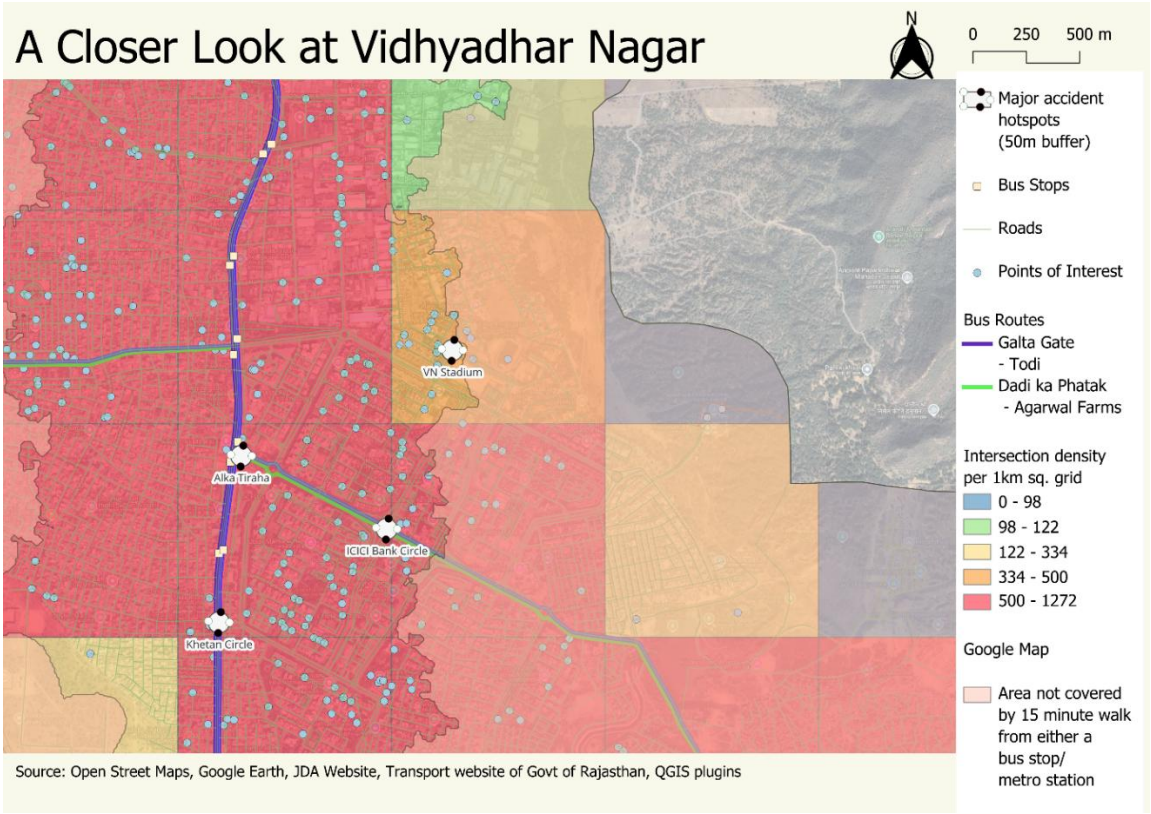
- Most of the city is in the high intersection zone.
- High accident hotspots overlap at dense intersections.
- Higher intersection density does not always correlate to higher pedestrian safety.

density of 500-1272 areas (dark red). For instance, 4 accident hotspots are in Vidhyadhar Nagar, with 3 of them within a single 1km square grid of 926 intersection points, far exceeding the ideal range. Though higher intersection points mean the area is potentially walkable due to higher connectivity, excessive density could also mean chaotic junctions limiting pedestrian movement. This coupled with accident hotspots thus becomes a paradox highlighting impaired walkability. This observation warrants a closer look at the area.

### 5.1 Urban Anatomy of Vidhyadhar Nagar

Vidhyadhar Nagar was developed as a very well-planned township, in the northern part of the city, blending ancient Indian town planning principles with urban design features. Makkar and Gangwar (2023) highlight that this neighbourhood does create a pedestrian-accessible zone with a walkable core, elevated footways, and mixed-use development. However, green spaces are located on the periphery of the neighbourhood where there also exists a lack of commercial amenities. This violates the "five-minute walking" principle[15] as it impels people to use vehicles to access such amenities. Also, residents from the core or the "central spine" have to travel via high-speed roads to access green open spaces.

[15] Makkar and Gangwar (2023) mention this principle which is adopted in the walled city area of Jaipur where all commercial activities are accessible within a 400-500m distance.




Map 4: A Closer Look at Vidhyadhar Nagar

Key takeaways:

- The neighbourhood hosts 4 major accident hotspots and lies in the very high intersection density zone.
- Last-mile connectivity in peripheral areas is limited.
- There is a preference for walking in the area and thus supporting infrastructure for the same should be created.

Though the area is highly accessible with respect to bus routes, a survey (Makkar and Gangwar, 2023) finds that only around 10% usage of public transport. It also reveals that improper pedestrian crossings results in traffic conflicts. This coupled with the accident hotspots demonstrates a breakdown in pedestrian safety infrastructure with vehicular dominance. This could also suggest that there exists a poor intersection design such as wide crossings. The survey also highlights that close to 80% respondents find the neighbourhood safe and around 40% prefer to walk, thus necessitating the need for enhanced walkability, especially on the peripheral area. The eastern part of the peripheral area also represents the area not covered within a 15-minute walk radius from either a bus stop or a metro station, thus limiting practical walkability. The western part is highly accessible through bus. The survey findings also underscore this issue wherein though nearly 40% of residents prefer walking and have rated their comfort level high, nearly half of them are relying on two-wheelers. This highlights the need to improve intersection design for better walkability as well as reduction in vehicular

accidents. Improving poor peripheral connectivity and integrating green and commercial zones within walking distance could be the next viable step.

### 5.2 High Access and High-Risk Zones

There are similar instances of highly accessible areas whilst also being unsafe due to presence of multiple accident hotspots (refer Map 4). One of them is the Delhi Highway area (from RAC cut to Eidgah Mandi) where 4 accident hotspots fall in the very high intersection density, with 600 to 770 intersections. This area nearly falls in the 15-minute walk radius from a bus stop. Similarly, the area around Malviya Nagar and Tonk Road hosts 5 accident hotspots and 4 of them fall the very high intersection density, with around 600 intersections. Another area is the 200ft Bypass near Nirman Nagar wherein 3 accident hotspots fall in the very high intersection density. These areas are heavily bus accessible with multiple bus stops.

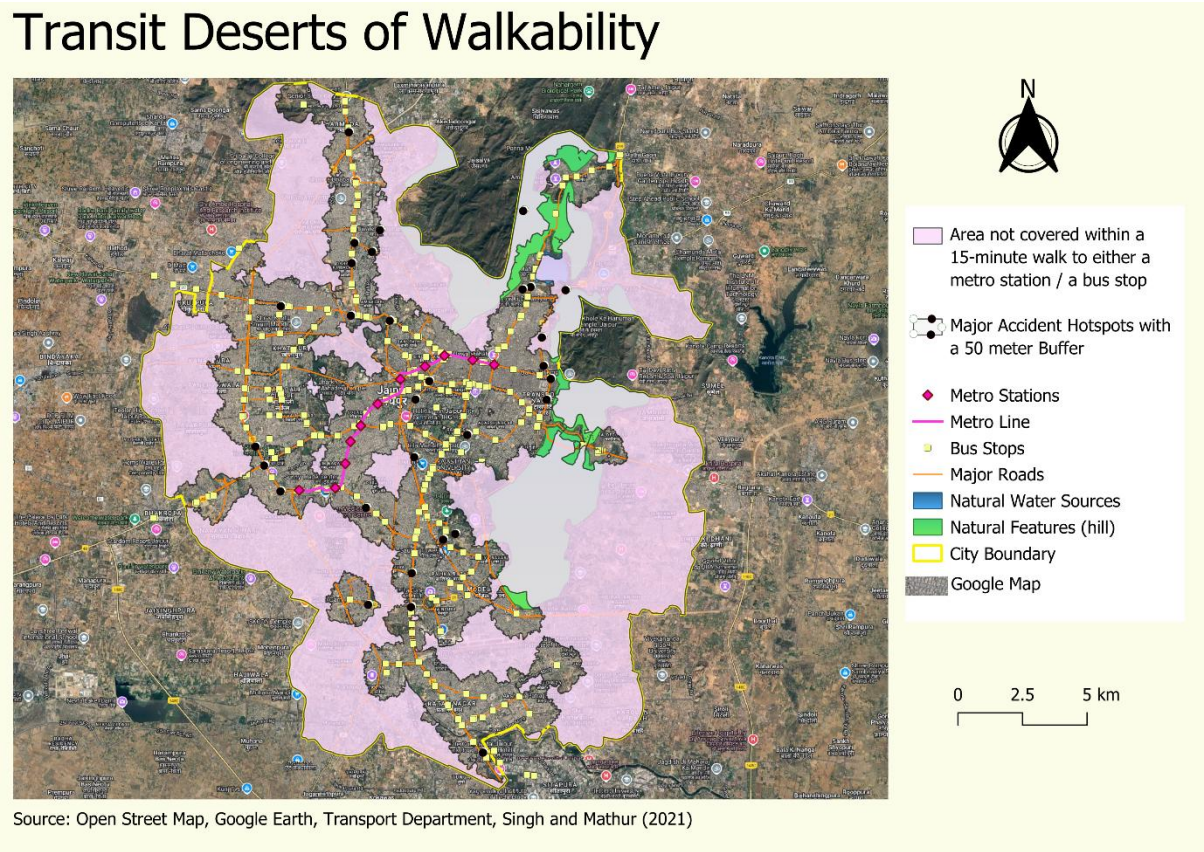
It is to note that all the accident hotspots clusters fall in the very high intersection density area and none of the accident hotspots fall in the ideal range intersection density range of 98 to 122 (green). Thus, very high intersection density does not always translate to pedestrian safety. These transit-rich zones with multiple transit nodes should ideally facilitate modal shift to walking and public transport, however the environment remains unsafe. However, this could be due to the fact that majority of the city is in the range of very high intersection points. Nonetheless, intersection density is meaningless without context and actual street situation, traffic conflicts and accessible amenities should also be in focus. Comfort in theory does not equate to safety in practice. It is not to imply to cap intersection count but to reimagine walkability with safety. This entails reclassifying unsafe intersections for urgent redesign with traffic calming measures.

## 6. Transit Deserts of Walkability

The transit deserts in Map 5 highlight the areas which not accessible to residents within a 15-minute walking distance from public transportation transit point – either a metro station or a bust stop. Jaipur's current metro line runs majorly in the central zone of the city, thus neglecting all other parts of the city. The metro is used for only 0.1% of the total city trips and only 2.5% of the population lives within a 500-metres of a metro station (UN Habitat, 2021). Though the second phase of the metro aims to broaden reach and accessibility, the first phase itself grapples with persistent issues of low ridership. Moreover, there has been little to no tangible advancement to date on the development of the second phase. As for bus stops, they concentrate on major roads like Tonk Road and Ajmer Road, however neglect other peripheral areas of the city.

In the south these areas of transit gaps are Muhana Road and Sanganer with industrial transition and peri-urban transition. In the west, there are expanding suburbs like Narayan Vihar, Mansoravar extension, Lalarpura and Panchyawala where there are infrastructure gaps with incomplete road networks. These are fast-growing suburbs with unplanned housing

developments. In the north, there are mixed-use corridors like Loha Mandi (industrial), Dadi ka Phatak (transport), VKI area (industrial/commercial) and Shastri Nagar (residential). These are traffic bottlenecks which also surround a couple of major accident hotspots. In the west, areas of transit gaps such as Jaisinghpura, Jamdoli and Luniyawas are on the rural-urban fringe wherein farmlands coexist with informal settlements coupled with unpaved roads for walking and limited access to other facilities. Thus, these transit deserts are shaped by spatial inequalities and land-use disparities and require tailored interventions to enhance walkability and accessibility. JDA's Comprehensive Traffic and Transport Study had aimed to close this gap by providing access to 60% of the population to a public transport facility within a 500-metre walk by 2024-25 (HT Correspondent, 2019). However, actual progress is yet to be confirmed.




Map 5: Transit Deserts of Walkability

Apart from the peripheral areas, there are certain central regions of the city that also lack transit coverage, especially area surrounding Imli Phatak and Karatarpura Phatak which is a dense lower to middle-income residential area. Dhanka Basti and portion of Hasanpura reflect informal housing coupled with low accessibility. In addition to being a transit desert, pedestrian infrastructure is virtually absent. Purani Basti and Mohan Nagar in the walled city suffer from a disconnect between physical proximity to stations and accessing them via foot. Though they are geographically close to transit points like Badi Chaupar, walkability is hindered due to non-existent footpaths, encroachments, and high congestion during peak traffic hours. Jhalana Doongri and JDA Colony in the southeast have spatial fragmentation. Most of these areas are population dense and thus undermines effectiveness of expensive infrastructure like metros. Kakkar et. al. (2020) highlight that majority of the transport policies in the country fail to capture mobility inequalities. Thus, it becomes imperative to have renewed focus on these peripheral areas. Expanding and decentralizing bus network and creating integrated micro-mobility hubs, with first/last-mile incentives from such hubs can address the issue. Additionally, pedestrian mobility and integrated access to nearby transit stops should be focussed on in high-density informal settlements like Dhanka Basti.

Map 5 highlights that large parts of peripheral Jaipur remain transit deserts, due to spatial or infrastructural inequalities. This neglect aggravates poor walkability.

## 7. Social Equity: Reclaiming the Streets

Urban planners across the globe have increasingly come to the realisation that cars act as barriers to pedestrian-friendly cities (Laker, 2017). Urban design today is car-centric with driving even for short trips. Speck has highlighted that the post 19th century separation of land use coupled with oversized infrastructure like wide streets and excessive lanes has induced a car-dependent urban sprawl (TEDx Talks, 2014). The suburban sprawl today is characterized by design. Kannan (2024) highlights that Indian cities already possess a mixed-use design model however; the lack of non-motorised transport infrastructure pushes them towards being pedestrian unfriendly. The irony of the car culture is emphasised in the fact that people drive fairly long distances to walk in a park, like to the Central Park in Jaipur (seen as the city's central lung).

This reinforces the need of prioritizing human-centric design rather than vehicular convenience. Mario Alves of International Federation of Pedestrian further reiterates that the management of cars is the answer to enhancing walkability in cities, since lesser the traffic, more the safety (Laker, 2017). Pollution – both air and noise – deters walking and takes away the appeal of walking as a routine activity or even leisurely. Globally cities like Oslo, Madrid, Helsinki, and Barcelona have encouraged car-free zones and plazas in some form which has

led to people reclaiming the public space, inhabit the streets and making walking a default choice rather than the last resort.

Only 14.4% of the total road length in the city have more than 1.2 metres wide footpaths and only 35% of the total population has a public transportation stop with a 500-metre distance (UN Habitat, 2021). This spatial neglect, in light of vehicular dominance, does not only signal at poor planning but also reflects a hierarchy of mobility. People without private transportation are thus denied access to the public realm as well. It has thus become a case of technology determinism where a technology (automobiles) has restructured urban space with respect to how people live, work and interact (public transit, walking). Social life is reorganized around this technological choice and public transportation as a public good becomes underutilised – as a residual or fallback for poor. Legal mandate for walkability audits and enforcement of pedestrian rights across all mobility projects will ensure inclusive public spaces for all users.

### 7.1 Streets for Whom?

Datey et. al. (2012) underscores that for Indian roads to be truly "inclusive" they should accommodate pedestrians, cyclists, rickshaws, and vendors along with vehicles. Vendors are often viewed as nuisances which encroach public spaces and thus have on-going tension between them and authorities. However, beyond their economic role they also provide affordability and accessibility, which are absent in formal markets. Another key contribution street vendors and hawkers play is by serving as "eyes on the street"[16] – a concept by Jane Jacobs[17] that elaborates that vibrant streets with continuous activity creates natural informal surveillance. One of the key reasons for encroachment of vendors on the streets is the absence of any policy or guidelines in India that integrate vendors into pedestrian infrastructure design (Singh and Dev, 2023). Thus, thoughtful integration with regulated vending can enhance pedestrian access and safety.

With respect to women, this discussion becomes even more vital as UN Habitat (2021) highlights that, for working women who travel to work, walking is the most common commuting method (37%), followed by bus transit (25%). Spatial policing becomes evident as missing footpaths push women to self-regulate their routes and schedules. Perceptions of "unsafe" streets with notions of "acceptable" hours also dictate unwritten dress codes. These restrictions are further compounded with the time-poverty women face. The burden of unpaid care work compels women to engage in trip-chaining (UN Habitat, 2021). Women combine multiple errands – taking children to school, going to work, buying grocery, taking an elder to a clinic, etc. – into one journey, making it a complex journey, as opposed to the existing linear, peak-hour commutes in the transport system. In reimagining urban mobility, these insights are crucial towards building mobility infrastructure for equitable cities.

[16] Jagannath (2016)
[17] An American-Canadian journalist and author of "The Death and Life of Great American Cities" among her other books.

## References


Baobeid, A., Koç, M., & Al-Ghamdi, S. G. (2021). Walkability and its relationships with health, sustainability, and livability: elements of physical environment and evaluation frameworks. *Frontiers in Built Environment*, *7*. https://doi.org/10.3389/fbuil.2021.721218

*Comprehensive Mobility Plan (CMP) May, 2018*. (n.d.). https://transport.rajasthan.gov.in/content/transportportal/en/metro/Project/comprehensive-mobility-plan--cmp--may--2018.html

Correspondent, H. (2019, March 4). 60% Jaipur population to have public transport access within walkable distance in 5 years. *Hindustan Times*. https://www.hindustantimes.com/jaipur/60-jaipur-population-to-have-public-transport-access-within-walkable-distance-in-5-years/story-mHnPpOpYep6YXRgb1OkE7L.html

Datey, Abhijit, et al. "Walking and Cycling in Indian Cities: A Struggle for Reclaiming Road Edges." *Centre for Urban Equity*, Nov. 2012. *Research Gate*, https://cept.ac.in/UserFiles/File/CUE/Working%20Papers/Revised%20New/18CUEWP18_Walking%20and%20Cycling%20in%20Indian%20Cities%20A%20Struggle%20for%20Reclaiming%20Road%20Edges.pdf.

Ewing, R., & Cervero, R. (2010). Travel and the Built Environment: A Meta-Analysis. *Journal of the American Planning Association*, *76*(3), 265–294. https://doi.org/10.1080/01944361003766766

Gadru, Shikha. "Pedestrian Movement and Visual Orientation in Public Spaces." *Journal for Studies in Management and Planning*, vol. 03, no. 01, Jan. 2017, pp. 55–62. *Academia.edu*, https://d1wqtxts1xzle7.cloudfront.net/51901767/Pedestrian_Movement_and_Visual_Orientation_in_Public_Spaces-libre.pdf?1487789904=&response-content-disposition=inline%3B+filename%3DPedestrian_Movement_and_Visual_Orientati.pdf&Expires=1745999894&Signature=IZbSv2qC-z2UwgSVsStzmgeC65Fhh6KCL~mDq9xTyhIODK33fYX2mAFlqCmuaUj86xo6NFPacZGNojK5hCsrZwCgIKAi8q8lI--mcDuiHpUahfGhhfNtL6EiLK09dltvhTuL1PaXd5j~IATiILt5CePSAOjMnxMPog3Ad4AIc-HcKQN9vF0vEIoe8CGT2XuSmdK7MxL870~-SiAGUpj~KAFjQjZ2nC9kF72rjElLhBbum30qUOPT78aSn2Rz2afxLoZfiEgtYujCJedciKhMNcCVjmoo7smkz3VHs0KxlvkhqHwTpkSIa0bqm1F9FJ~vqqVUtMcr19s1SW24bYvq1w__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

Hazarika, Arshia Khajooria, et al. "Urban planning: An ecological and cultural reflection , Urban History as a Planning Tool." *APSA congrees 2011-TOKYO*, ResearchGate, 2011, www.researchgate.net/publication/301956498_Urban_planning_An_ecological_and_cultural_reflection_Urban_History_as_a_Planning_Tool.

"How Indian Cities Failed Public Transport | A Quint Deep Dive." *YouTube*, directed by The Quint, 26 Dec. 2024, youtu.be/JkNLUZa5INk?si=iAH0iXNvQgfAbjpg. Accessed 28 Mar. 2025.

Kakar, I. S., Peden, M., & Jagnoor, J. (2020). Intersectionality based policy analysis: Equity in mobility in India. *Transport Policy*, *101*, 14–22. https://doi.org/10.1016/j.tranpol.2020.11.010

Kannan, Sowmya. "15-minute cities." *Citizen Consumer and Civic Action Group (CAG)*, 18 Feb. 2024, www.cag.org.in/blogs/15-minute-cities. Accessed 11 Apr. 2025.

Kumar, M., Singh, S., Ghate, A. T., Pal, S., & Wilson, S. A. (2016). Informal public transport modes in India: A case study of five city regions. *IATSS Research*, *39*(2), 102–109. https://doi.org/10.1016/j.iatssr.2016.01.001

Laker, Laura. “Where is the world’s most walkable city?” *The Guardian*, 12 Sept. 2017, amp.theguardian.com/cities/2017/sep/12/walkable-city-worlds-most-new-york-melbourne-fes-el-bali.

Makkar, Yogesh, and Gaurav Gangwar. “Critical Evaluation of Ancient Indian Town Planning Principles Used in Neighborhood Design: A Case Example of Vidyadhar Nagar.” *Journal of Civil Engineering and Environmental Technology*, vol. 10, no. 1, Jan. 2023, pp. 1–11. https://d1wqtxts1xzle7.cloudfront.net/101012087/10Apr2023100435Ar._Yogesh_Makkar-libre.pdf?1681301499=&response-content-disposition=inline%3B+filename%3DCritical_Evaluation_of_Ancient_Indian_To.pdf&Expires=1745055544&Signature=VRsnP8-ZpBxDqIBRGoSfuCKTjmN2RZ81hJR~nqk~eo7X-R090Kkl1iwb9QLgoszuShLFVxhp7h0lwC9caCfvAPrnep~h0OiL0KYx7dI-UqsvcjwLbl87swTcUVzhJ06gX551dtGrTgHZVU-XziMfn1j8Ahf6o8HrKsXyq35JmXOzEpaHthMIIak4tRmVD1nmSpbWEP9zvIOitJRIgQTN-D0hvZC3H39Md5iPXF300vEHHcgR0wJGsxk~lSmEsOHMNCdv3Mz1Qg12tAzIafPIgPXtroj9CiXKX8jfy818MM6C53SqNs89dd-FARYl~b2EdKMS2M2RqEvO0N8IgeU82w__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

Mehta, Vikas, and Jennifer K. Bosson. “Third Places and the Social Life of Streets.” *Environment and Behavior*, Nov. 2010. *Research Gate*, https://doi.org/10.1177/0013916509344677.

Oldenburg, Ray, and Karen Christensen. “Third Places, True Citizen Spaces.” *The UNESCO Courier*, 23 Mar. 2023, https://courier.unesco.org/en/articles/third-places-true-citizen-spaces#:~:text=In%20the%201980s%2C%20American%20sociologist,example%20of%20the%20third%20place. Accessed 2 May 2025.

OMI Foundation. (2023, April). Ease of Moving Index - India Report 2022 https://easeofmoving.org/wp-content/uploads/2023/11/easeofmoving-2022.pdf#page=133.09

Pareek, Gaurav, and Deepak Mathur. “Comprehensive Review and Analysis of Urban Transport Challenges and Sustainable Transport Framework for Jaipur City.” *International Journal of Civil Engineering Applications Research (IJCEAR)*, vol. 3, no. 1, 2022, pp. 1–14. www.technicaljournals.org/IJCEAR/index.php/journal/article/view/1/1.

Practical Cities. “Masterclass 27 ‘QGIS in Urban Transportation and Mobility’ by Ujaval Gandhi (Spatial Thoughts).” *YouTube*, 30 Apr. 2021, www.youtube.com/watch?v=ihc9Xa8a7LU. Accessed 20 Mar. 2025.

Pipralia, Satish, et al. “Exploring the Imageability of Urban Form in Walled City Jaipur.” *Annual International Conference on Architecture and Civil Engineering (ACE 2016)*, Research Gate, 2016, https://doi.org/10.5176/2301-394X_ACE16.43.

Sallis, J. F., Cerin, E., Conway, T. L., Adams, M. A., Frank, L. D., Pratt, M., ... & Owen, N. (2016). Physical activity in relation to urban environments in 14 cities worldwide: a cross-sectional study. The lancet, 387(10034), 2207-2217.

Sharma, A., Garg, S., Shah, V., & Lohan, P. (2015). BASICS OF DESIGN: LESSONS FROM WALLED CITY OF JAIPUR. *ResearchGate*. https://www.researchgate.net/publication/286249968_BASICS_OF_DESIGN_LESSONS_FROM_WALLED_CITY_OF_JAIPUR

Singh, A. S., & Dev, M. (2023). Challenges of successful non-motorized transport infrastructure in Indian cities. *Transportation Research Procedia*, *72*, 1982–1989. https://doi.org/10.1016/j.trpro.2023.11.679

Singh, Jitendra, and Deepak Mathur. “A Statistical Data Analysis of Road Accidents and Traffic Volume in Jaipur.” *International Journal of Recent Research and Review*, vol. 14, no. 3, Sept. 2021. www.ijrrr.com/papers14-3/paper3-A%20Statistical%20Data%20Analysis%20of%20Road%20Accidents%20and%20Traffic%20Volume%20in%20Jaipur.pdf.

TEDx Talks. (2014, May 23). *The general theory of walkability | Jeff Speck | TEDxMidAtlantic* [Video]. YouTube. https://www.youtube.com/watch?v=uEkgM9P2C5U

UN Habitat. “Jaipur: City Profile and Diagnostic Report: Sustainable Cities Integrated Approach Pilot (SCIAP).” *UN Habitat*, Aug. 2021, www.unhabitat.org.in/sites/default/files/2024-11/city_profile_and_diagnostic_report_jaipur.pdf. Accessed 30 Apr. 2025.

Upadhyaya, Vibha, and Jakhanwal. “Traditional Walled Cities of Rajasthan India: A Sustainable Planning Concept.” *International Journal of Engineering Research & Management Technology*, vol. 2, no. 3, May 2015. *IJERMT*, ijermt.org/publication/15/IJERMT%20V-2-3-32.pdf.

Vashishtha, Abhishek. “Impacts of Transportation Infrastructure on Urban Air Quality in Jaipur Metro City.” *International Journal of Advanced Research in Arts, Science, Engineering & Management (IJARASEM)*, vol. 10, no. 1, Jan. 2023, ijarasem.com/admin/img/54_Impacts.pdf.

*What are the Gestalt Principles?* (2024, December 7). The Interaction Design Foundation. https://www.interaction-design.org/literature/topics/gestalt-principles